Universidad Internacional del Ecuador (UIDE)

Computer Science School

Network Communication Technology

Title: *GTA1_Phase_1_Integrated Project*

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1. Scenario and Scope

Due to the organization's expansion, it has two main headquarters (Headquarters A and Headquarters B). To resolve information loss and delays caused by the use of informal channels, the organization needs an enterprise platform responsible for managing internal corporate emails, ensuring that communication flow is recorded. Currently, our scope is limited to planning, logical-physical design, VLSM addressing strategy, and network architecture modeling for both headquarters, ensuring that communication and information are transported efficiently and securely.

Headquarters Structure:

Both headquarters share the same network architecture:

- An OpenWrt-based edge router (on Raspberry Pi) with independent Internet access, responsible for firewall rules, NAT, DHCP services, and dynamic routing.
- A redundant switching system consisting of two Cisco switches interconnected via EtherChannel (LACP/PAgP) and protected against loops using RSTP.

Estimated Devices per Segment (VLANs):

To perform VLSM addressing over the corporate block 10.10.0.0/16, the following devices are estimated per headquarters:

- VLAN 10 (Users / Operations): Estimated 30 end devices per headquarters.
- VLAN 20 (Servers): Estimated 4 devices per headquarters.
- VLAN 30 (Guests): Estimated 15 temporary devices per headquarters.
- VLAN 99 (Management): Estimated 4 devices per headquarters.

Communication Needs:

- Inter-headquarters Communication: It is required to interconnect the two headquarters through a shared Wi-Fi link (192.168.100.0/24) using the OSPF dynamic routing protocol, ensuring fast convergence and low latency for corporate network traffic.

- Client-Server Traffic: Operational users located in VLAN 10 of either headquarters must be able to send TCP/HTTP requests to the internal mail backend located in VLAN 20, mandatory validating their session token via FastAPI before executing any transaction.
- Isolation and Security: Strict isolation via VLANs and Firewall rules on OpenWrt, preventing guest VLAN traffic or external networks from compromising the critical server and management infrastructure.

2. NETWORK REQUIREMENTS

- **OpenWrt Edge Routers on Raspberry Pi:** An OpenWrt-based edge router running on a Raspberry Pi will be deployed at each headquarters (Raspberry 1 for Headquarters A and Raspberry 2 for Headquarters B) to manage routing and edge operations. The WAN interface (wlan0) will operate over the shared wireless segment 192.168.100.0/24, assigning 192.168.100.10/24 to Headquarters A and 192.168.100.11/24 to Headquarters B, with gateway 192.168.100.1/24. The LAN interface (eth0) will function as an IEEE 802.1Q tagged trunk link connecting to the first switch at each headquarters.
  The edge routers will manage firewall rules, NAT, DHCP services, and dynamic routing with independent Internet access.
- **Segmentation and Trunks with IEEE 802.1Q Standard:** The physical network infrastructure will be divided into four distinct broadcast domains per site, segmenting operational staff into VLAN 10, internal servers into VLAN 20, guest visitors into VLAN 30, and technical management into VLAN 99. All switches and OpenWrt routers will be interconnected via trunk lines configured under the IEEE 802.1Q frame tagging standard. This mechanism ensures that Ethernet frames maintain strict logical separation across the switched fabric, preventing unauthorized frame leakage across layer two boundaries.
- **VLSM Addressing Strategy:** The logical addressing design will be carved out from the assigned corporate class block 10.10.0.0/16 using Variable Length Subnet Masking. Each subnet will be dimensioned strictly based on the estimated endpoint density per segment, selecting subnet masks that maximize address utilization and eliminate waste. To ensure deterministic dynamic routing

between Headquarters A and Headquarters B, each headquarters will be provisioned with dedicated, non-overlapping subnet blocks.

- **Inter-VLAN Routing via Router-on-a-Stick:** Traffic forwarding between distinct virtual local area networks will be handled via a Router-on-a-Stick topology centered on the OpenWrt platform. Logical subinterfaces associated with each VLAN identifier will be created on the physical LAN port of the edge router. Each subinterface will act as the default gateway for its respective broadcast domain, enforcing policy inspection before routing packets between internal corporate segments.
- **High Availability Switching with EtherChannel and RSTP:** The Cisco distribution and access switching layer at each headquarters will provide physical link redundancy and fault mitigation through EtherChannel aggregation running the standard LACP protocol. Additionally, Rapid Spanning Tree Protocol will be enabled across the entire switched topology. This architecture aggregates bandwidth between switches and neutralizes layer two switching loops, providing instant sub-second failover in the event of interface or physical link disconnections.
- **Dynamic Routing with OSPFv2:** Inter-headquarters routing will be established using dynamic OSPFv2 routing within Area 0 over the shared wireless transport link. The OpenWrt routers will automatically exchange internal routing tables to establish reliable, bidirectional communication between both facilities. To preserve independent Internet breakout for each headquarters, edge routers will refrain from originating or advertising the default route into the OSPF domain.
- **Network Services and Perimeter Security:** Each OpenWrt router will run dedicated DHCP server scopes to dynamically deliver IP addressing, netmasks, and gateway settings to client endpoints on each VLAN. External connectivity will be handled through NAT masquerading applied to the outbound wireless WAN interface at each facility. Furthermore, stateful perimeter firewall rules will enforce complete isolation on the guest VLAN, allowing only outbound Internet transit while denying any access toward corporate servers or management infrastructure.

3. Preliminary Topology

Physical Topology

The physical infrastructure of each headquarters is designed with high availability in the access/distribution layer and standardized edge equipment:

Edge Routers (OpenWrt):

- A Raspberry Pi (or equivalent platform) with OpenWrt is used at each headquarters (Raspberry 1 for Headquarters A and Raspberry 2 for Headquarters B).
- WAN Interface (wlan0): Connected via a shared Wi-Fi/WAN segment (192.168.100.0/24) managed by a common AP/Gateway (192.168.100.1/24). The assigned IP addresses are 192.168.100.10/24 for Headquarters A and 192.168.100.11/24 for Headquarters B.
- LAN Interface (eth0): Functions as a tagged trunk link under the IEEE 802.1Q standard toward the headquarters' first switch.

Cisco Switches (Access and Distribution Layer):

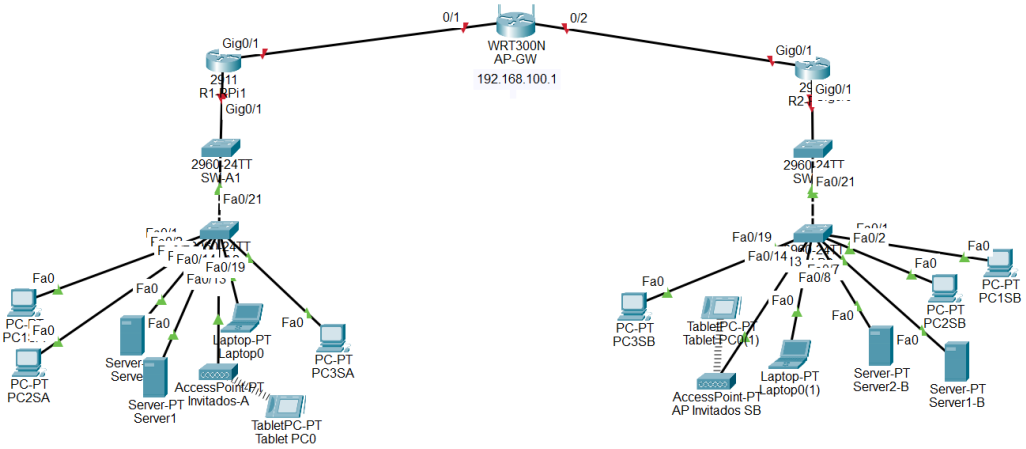
- Each headquarters has a pair of Cisco switches (SW-A1 / SW-A2 for Headquarters A and SW-B1 / SW-B2 for Headquarters B).
- EtherChannel (LACP): The switches are interconnected via high-speed aggregated links using LACP for redundancy and increased bandwidth.
- RSTP (Rapid Spanning Tree Protocol): Enabled in the switch topology to prevent Layer 2 loops in the event of multiple physical routes.

End Devices

Connected to the access ports of the switches according to their respective VLAN (user computers, servers, guest mobile devices, and management equipment).

Figure 1

Physical topology design



Note. Own elaboration

Logical Topology:

The logical architecture defines traffic segmentation and routing between the different broadcast domains:

Addressing Plan (VLSM):

- It starts from the assigned main institutional block: 10.10.0.0/16.
- This block is divided using Variable Length Subnet Masks (VLSM) to optimize address space in each subnet.

VLAN Segmentation:

- VLAN 10: Users / Workstations (10.10.X.0/Y).
- VLAN 20: Servers (Go Application, FastAPI, databases) (10.10.X.0/Y).
- VLAN 30: Guest Network (10.10.X.0/Y).
- VLAN 99: Network management and infrastructure (10.10.X.0/Y).

Inter-VLAN Routing (Router-on-a-Stick):

The OpenWrt router handles tagged subinterfaces (802.1Q) on its LAN port to route traffic between the different VLANs in a centralized manner.

Dynamic Routing (OSPFv2):

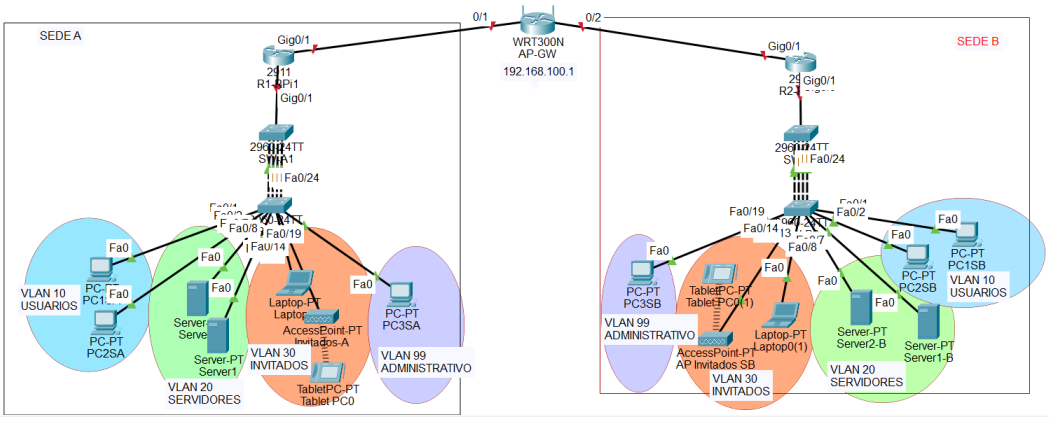
- OSPFv2 (Area 0) is implemented over the shared wireless/WAN network segment (192.168.100.0/24).
- The OpenWrt routers dynamically exchange the routes of their respective local subnets, allowing Site A to communicate transparently with Site B. The default route is not advertised through OSPF to keep Internet egress independent for each site.

Edge Services in OpenWrt:

- NAT: Enabled on the WAN interface to allow Internet access from the internal networks of each site.
- DHCP: DHCP servers configured per VLAN on the OpenWrt router.
- Firewall: Access control rules to protect sensitive resources and restrict unauthorized traffic between zones.

Figure 2

Logical topology design



Note. Own elaboration

4. ADDRESSING STRATEGY

To comply with segmentation requirements, the distribution of the 10.10.0.0/16 IP address block will be carried out using VLSM subnet masks. Addressing will be performed as follows:

- **VLAN 10 (Users):** Taking into account the 30 devices per site corresponding to administrative, sales, and operational personnel, a subnet will be assigned using the appropriate VLSM prefix to cover the volume of end-user equipment without generating IP address waste.
- **VLAN 20 (Servers):** Taking into account the 4 devices per site responsible for hosting the Go client-server application, the FastAPI authentication microservice, and database services, a strict, protected, smaller segment will be sized to guarantee exclusive addressing for the internal services infrastructure.
- **VLAN 30 (Guests):** Taking into account the 15 temporary devices per site, a specific subnet will be calculated to support external visitor concurrency, restricting their connectivity through routing policies that direct their traffic exclusively to Internet egress.
- **VLAN 99 (Management):** This VLAN is intended for 4 devices per site allocated for secure SSH management console access and administrator terminals; a highly restricted segment will be reserved to protect critical network control points and servers.

5. INTEGRATION PLAN

- **Software Component Placement and Architecture:** The distributed enterprise software components will be deployed within the secured VLAN 20 server subnet across both headquarters to shield core application logic from untrusted network access. This zone will host the client-server corporate messaging engine built in Go, which manages concurrency via goroutines and handles message persistence. The FastAPI authentication microservice will reside in the same server segment to centralize credential verification and token signing. Operational users will access the platform exclusively from client endpoints residing in VLAN 10.
- **Transactional Flow and Inter-Headquarters Communication:** The end-to-end communication sequence begins when an operational workstation in VLAN 10 transmits an authentication request to the FastAPI service located in VLAN 20. Once credentials are verified, the authentication service returns a cryptographically signed session token to the requesting client. The user workstation then attaches this token to the authorization header of outbound

messaging requests directed to the Go backend server. Requests bound for the alternate headquarters will be routed across the OpenWrt WAN interfaces over the OSPF wireless link without manual intervention. Before persisting data or processing transactions, the Go backend will validate the cryptographic session token against the FastAPI service, ensuring that unauthenticated actions are rejected.

- **Access Control Policies and Port Filtering:** The OpenWrt stateful firewall at each headquarters will serve as the primary policy enforcement barrier between network segments. The firewall will permit traffic from VLAN 10 to VLAN 20 targeted exclusively at TCP port 8000 for FastAPI authentication requests and TCP port 8080 for transactions directed to the Go mail service. Communication between application backends and relational database engines will remain strictly confined within the boundaries of VLAN 20. In contrast, all incoming traffic originating from the guest VLAN 30 directed toward corporate subnets will be dropped immediately. Remote administrative access via SSH will be restricted exclusively to verified endpoints located within management VLAN 99.
- **Platform Resilience, Auditing, and Verification:** Any network transaction directed toward the application server that lacks a valid session token or contains expired credentials will be instantly dropped with an unauthorized status response, shielding computational resources from abuse. System resilience will be upheld through the combined layer two high-availability mechanisms of EtherChannel and RSTP, coupled with the fast convergence of OSPF across the shared wireless segment. Every transaction processed by the messaging engine will generate an audit log capturing sender details, origin IP address, and an exact timestamp, fulfilling the project mandate for formal, traceable corporate communications.

6. WORK PLAN

To ensure a structured, controlled, and traceable execution of the distributed enterprise platform, the development of communications and network infrastructure is governed by a 16-week schedule. Below is the detailed Gantt chart, which breaks down the activities, deliverables, and phases from initial planning and requirements analysis

to switch configuration, OSPF routing, integration testing, and final delivery. The attached legend indicates the progress status of each planned milestone.

GANTT CHART

PHASE	ACTIVITIES	WEEKS															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
PHASE 1: CONTEXT, PLANNING, AND ORGANIZATION	Initial Git repository with a README and license																
	Short video introducing the project and assigning roles																
PHASE 2: ANALYSIS AND REQUIREMENTS	General Objective and Specific Objectives by Course																
	Requirements Specification with Acceptance Criteria																
	Initial Architectural Design																
	Use-Case Diagrams and Flow Descriptions																
	Initial Traceability Matrix																
	Preliminary Sizing of VLANs, Services, and Addressing Requirements																
	Explanatory Video on Analysis and Scope																
PHASE 3: SOLUTION DESIGN	Logical and Physical Architecture of the Two Campuses																
	VLSM Plan, Routing Table, VLANs, and Ports																
	Trunk Design, EtherChannel, RSTP, Router-on-a-Stick, and OSPF																
	Comprehensive Architecture Diagram and VLSM/VLAN Plan																
PHASE 4: PROTOTYPE AND INITIAL DEVELOPMENT	Initial Configuration of Switches, VLANs, Trunks, EtherChannels, RSTP, and OpenWrt																
	Updated repository and modular structure with supported configurations																
	VLAN, trunk, EtherChannel, RSTP, and OpenWrt are running in the lab																
	Teaser video and partial demonstration																
PHASE 5: INTEGRATION AND DEPLOYMENT	Final OSPF Configuration and Route Verification																
	Implementation of NAT, DHCP, and Firewall Rules																
	Communication Policies Between VLANs and Sites																
	Integration Testing and Troubleshooting																
	Backups of switch configurations and OpenWrt																
	Connectivity matrix and access policies																
	Demonstration of communication between locations																
PHASE 6: TESTING, VALIDATION, AND EVALUATION	Intra-VLAN, inter-VLAN, and site-to-site connectivity testing																
	Verification of OSPF, EtherChannel, RSTP, DHCP, NAT, and firewall																
	Test plan and test case matrix with results																
	Requirements compliance matrix																
	Final candidate version of the system and demonstration video																
PHASE 7: FINAL PRESENTATION AND DEFENSE	Live demonstration of the application across components or locations																
	Final document (Report)																
	Complete, tagged repository with a final version																
	Academic presentation																
	Final demonstration video and oral defense																
PHASE 8: CLOSING AND FINAL DELIVERY	Final Version of the Project																
	Record of Corrections Made																
	Installation, Configuration, and Implementation Manual																
	Self-Assessment and Peer Assessment																
	Lessons Learned and Suggestions for Improvement																

STATE LEGEND:

Completed Not started Not completed

Responsibility Assignment:

Table 1

Responsibility Assignment

Responsible	***Activity***
Pamela Toapanta	Design and implementation of the network infrastructure in both sites (OpenWrt routers, VLANs, OSPF, EtherChannel, RSTP, VLSM, DHCP, NAT, and Firewall).
Santiago Lucina	Object-oriented modeling, development of the client-server system in Go, persistence, concurrency control via goroutines, and integration with business logic.
Camila Paucar	Analysis of organizational context, system justification, stakeholder matrix, ethical considerations, data protection, and support in integration with the FastAPI microservice routes.

Note. Own elaboration.

7. RESOURCES AND RISKS

Implementation Environment:

The project is implemented in a distributed multi-site network environment consisting of two main headquarters (Headquarters A and Headquarters B). The core infrastructure is simulated and deployed using Cisco Packet Tracer and virtualized environments, interconnecting the facilities through a shared wireless/WAN network segment (192.168.100.0/24).

Hardware and Software

- **Hardware:**
  - **Edge Routers:** Raspberry Pi devices (or equivalent hardware) running OpenWrt at each headquarters.

  - **Switching Layer:** Pairs of Cisco 2960 switches per site configured with EtherChannel (LACP/PAgP) and Rapid Spanning Tree Protocol (RSTP).
  - **End Devices:** Workstations (VLAN 10), servers (VLAN 20), guest wireless devices (VLAN 30), and management terminals (VLAN 99).
- **Software:**
  - **Network OS & Simulation:** OpenWrt firewall/router firmware and Cisco Packet Tracer.
  - **Application Stack**: Go language for the internal mail backend system, FastAPI for session authentication and token verification, relational databases, and TCP/IP protocol stacks.

Technical Risks and Dependencies:

To ensure project continuity and the stability of the distributed platform, the following critical dependencies and technical risks associated with the network infrastructure have been identified:

Critical Dependencies:

- **OSPF Routing Synchronization and Convergence:** Successful, packet-loss-free communication between Site A and Site B relies strictly on the correct propagation of dynamic routes via OSPFv2 (Area 0) over the shared Wi-Fi/WAN segment (192.168.100.0/24). Any error regarding subnet masks or active/passive interfaces on the OpenWrt routers will disrupt inter-site connectivity.
- **Network Tagging Consistency (802.1Q):** The scheme's proper functioning depends on the VLANs (10, 20, 30, 99) being configured identically and synchronously on both the OpenWrt router LAN interfaces and the trunk links of the Cisco access switches.

Technical Risks and Mitigation Strategies

Shared Wireless Link Saturation / Unauthorized Traffic

- **Risk:** Since the physical WAN medium is a shared Wi-Fi segment between both sites, there is a risk of congestion caused by heavy traffic or intrusions on the guest network (VLAN 30) that could impact critical enterprise services.
- **Mitigation**: Strict access control policies will be implemented via the OpenWrt firewall, alongside rigid traffic segmentation by VLAN; this will prevent guest users or external traffic from saturating the bandwidth allocated to the server core (VLAN 20) and management (VLAN 99). In addition, NAT and perimeter prioritization rules will be applied.

8. VALIDATION PLAN

To verify the correct functioning of the network prior to its installation, the following tests will be carried out:

- **Connectivity and OSPF tests:** Use of basic commands such as ping, traceroute, and show ip ospf neighbor to check that the routes between Site A and Site B communicate correctly through the shared Wi-Fi link.
- **Redundancy and isolation verification:** Checking that EtherChannel and RSTP links operate without creating loops in the network, and validating that data traffic between different VLANs occurs solely through the routing configured on the OpenWrt routers.
- **Integration and security tests:** Checking that the system instantly blocks any access attempt lacking a valid session token, ensuring network protection and compliance with project objectives.

9. REPOSITORY LINK

- **GitHub Repository:** https://github.com/Slucina04060708/PROYECTO-INTEGRADOR